Project A - Operations Runbook

Purpose

This runbook describes the first response to the most likely production incidents for Project A. It is intended for the platform on-call engineer. Each section follows the same layout: symptoms, diagnosis, steps, escalation and prevention. The runbook does not replace judgement; if the situation does not match, escalate early.

General rules

• Declare the severity as soon as you know it and open the incident channel.

• Record every action with a timestamp in the channel.

• Prefer rollback over fixing forward during a severity 1 or 2 incident.

• Never change production data directly without a second engineer approving the change.

Incident 1: Elevated 5xx errors at the API gateway

Symptoms

The gateway error rate exceeds 2 percent for five minutes and the on-call engineer is paged.

Diagnosis

Open the gateway dashboard and group errors by upstream service. One upstream usually accounts for most failures.

Steps

1. Identify the failing upstream from the dashboard.

2. Check that service's pods with kubectl get pods and look for restarts or pending pods.

3. If a recent deployment is the cause, roll it back with the pipeline rollback job.

4. If pods are healthy, check the service's dependencies listed in the design specification.

Escalation

Escalate to the owning team's on-call if the error rate has not fallen within 15 minutes.

Prevention

Canary deployments to 10 percent of traffic would have limited the impact.

Incident 2: Spike in failed logins

Symptoms

The Identity Service reports more than 300 failed logins per minute.

Diagnosis

Compare the failure reasons. Many invalid_credentials from few IP ranges suggests credential stuffing; many service errors suggests a platform fault.

Steps

1. Check the audit search for the top source addresses.

2. If an attack is likely, enable the stricter gateway rule that limits login attempts to 3 per minute per address.

3. Notify the Information Security Office.

4. Do not disable the login endpoint; genuine customers must still be able to sign in.

Escalation

Page the Security Lead for any suspected attack.

Prevention

Add device reputation checks in a later release.

Incident 3: Statement queue backlog

Symptoms

Queue depth for statement requests exceeds 5000 for more than 30 minutes.

Diagnosis

Check worker pod count and look for a stuck job that is being retried.

Steps

1. Confirm the worker autoscaler is active; it should scale towards 20 replicas.

2. Look for jobs retried more than 3 times and move them to the parked queue.

3. If object storage returns errors, check the provider status page.

4. Tell Customer Operations the expected drain time so they can inform callers.

Escalation

Escalate to the Documents team if the backlog is still growing after an hour.

Prevention

Pre-scale workers on the last two working days of each month.

Incident 4: PostgreSQL replica lag

Symptoms

Replica lag exceeds 30 seconds.

Diagnosis

Run the lag query on the replica and check for long-running queries on the primary.

Steps

1. Identify queries running longer than 60 seconds on the primary.

2. Cancel any runaway reporting query.

3. If lag persists, remove the replica from the read pool by setting the pool weight to zero.

4. Rebuild the replica only if lag remains above 5 minutes.

Escalation

Escalate to the database administrators if the primary shows heavy write load.

Prevention

Move heavy reporting queries to a dedicated replica.

Incident 5: PostgreSQL primary failure

Symptoms

The primary database is unreachable and health checks have failed three times.

Diagnosis

Confirm the failure from two independent locations before acting.

Steps

1. Declare a severity 1 incident and open the incident channel.

2. Confirm that replica 1 is caught up using the replication status command.

3. To promote the replica, run promote-replica --confirm-region=eu-west-2 from the bastion host. The command refuses to run without the region flag.

4. Update the connection secret so services point at the new primary and restart the services in this order: Identity, Account, Transaction, Card.

5. Rebuild a new replica from the promoted primary once the incident is stable.

Escalation

The recovery time objective is 4 hours. If promotion fails, escalate to the database vendor support line.

Prevention

Run a failover rehearsal every quarter.

Incident 6: Redis memory pressure

Symptoms

Redis memory use exceeds 85 percent.

Diagnosis

List the largest key prefixes with the memory analysis tool.

Steps

1. Check whether the rate-limit prefix has grown unusually; this indicates an abusive client.

2. Remove expired session metadata keys older than the documented lifetime.

3. If memory stays above 85 percent, scale the Redis node to the next size during a low-traffic window.

Escalation

Escalate to Platform if evictions begin.

Prevention

Alert at 70 percent instead of 85 percent.

Incident 7: TLS certificate expiry warning

Symptoms

The certificate monitor reports less than 14 days to expiry.

Diagnosis

Identify which certificate is affected, public or internal.

Steps

1. For the public certificate, trigger the renewal job and verify the new certificate on the staging hostname.

2. For internal certificates, check that the certificate manager is healthy.

3. After renewal, confirm the gateway reloaded the certificate by checking the expiry date from outside the cluster.

Escalation

Escalate to Platform if renewal fails twice.

Prevention

The renewal job is now scheduled 30 days before expiry.

Incident 8: Notification dead-letter growth

Symptoms

The notif-dlq queue holds more than 200 messages.

Diagnosis

Sample messages from the queue and group them by channel and error.

Steps

1. If most failures are for one provider, check the provider status page.

2. Once the provider recovers, replay the queue using the replay tool with a rate limit of 50 messages per second.

3. Discard messages older than 24 hours, because they are no longer useful to customers.

Escalation

Notify the Engagement team.

Prevention

Add a second SMS provider for fallback.